

Jiarui Zhang 张家瑞

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Profile. I am a fourth-year undergraduate in the Yao Class at Tsinghua University, with a background in theoretical computer science and machine learning. My research spans randomized sampling, circuit complexity, and efficient machine learning systems.

EDUCATION

Tsinghua University, Beijing, China

Sep 2023 – Jun 2027 (expected)

B.Eng. in Computer Science and Technology, Institute for Interdisciplinary Information Sciences

GPA: 3.78/4.0 | TOEFL iBT: 5.5/6.0

Relevant Coursework: Algorithm Design (A-), Theory of Computation (A), Abstract Algebra (A), Natural Language Processing (A+)

AWARDS

- **National Olympiad in Informatics (NOI) 2021** – Gold Medal
- **NOI 2022 Guangdong Provincial Team Selection** – 1st Place

EXPERIENCE

Student Researcher

Aug 2026 – Present

Advised by Prof. Jingzhao Zhang (Tsinghua University)

[Blog post](#) ↗

- Proved tight exact-sampling round bounds for DLMs with polynomial-width, constant-precision, constant-depth Transformer backbones on regular-language input-output pair distributions: two rounds suffice with revision, whereas exact sampling for languages outside AC^0 requires $\Theta(\log n / \log \log n)$ rounds without revision.
- Constructed randomized- AC^0 trajectory samplers for fixed path-dyadic Markov chains.

Applied Research Intern

Jul 2025 – Sep 2025

Tencent CDG

Financial Language Models

- Designed model-based and prompt-driven reward scoring for a financial LLM, with evaluation focused on citation validity, factual accuracy, timeliness, and instruction following.
- Improved citation-format accuracy from 40% to 98%, content and timeliness accuracy from 78% to 97%, and IFEval accuracy from 38% to 80%.

Research Intern

Jan 2026 – Aug 2026

Binhang Yuan Research Group, HKUST

Efficient Machine Learning Systems

- Developed distributed training methods for Tree Attention, reducing memory usage and improving throughput for multi-turn LLM training.
- Contributed to dual diffusion draft models for D^2SD , a speculative-decoding method for accelerating LLM inference.

PUBLICATIONS

- **Jiarui Zhang***, Yuchen Yang*, Ran Yan*, Zhiyu Mei, Liyuan Zhang, Daifeng Li, Wei Fu, Jiaxuan Gao, Shusheng Xu, Yi Wu, Binhang Yuan. “AREAL-DTA: Dynamic Tree Attention for Efficient Reinforcement Learning of Large Language Models.” **ICML 2026**.
- Liyuan Zhang, **Jiarui Zhang**, Jinwei Yao, Ran Yan, Yuchen Yang, Jiahao Zhang, Tongkai Yang, Yi Wu, Binhang Yuan. “ D^2SD : Accelerating Speculative Decoding with Dual Diffusion Draft Models.” arXiv:2606.04446.

* Equal contribution.

INTERESTS

- Competitive Shuttlecock (毽球), member of the Tsinghua University Shuttlecock Team
- Hiking, especially single-day routes with $\approx 1,000$ m of elevation gain